

LINEAR ALGEBRA

Independent Study Assignments · Week 2

Topics: dot product, cross product, introductory matrices, matrix operations, and matrix-vector multiplication.

These problems form part of the course practice pool. Exam questions may use the same mathematical structure with different numerical values. Write vectors as columns and show enough work to explain your method.

You may work individually or discuss methods with classmates. If an AI tool helps clarify an idea, you must still perform and explain the calculation yourself.

Part A · Dot products, lengths, and angles

Exercise 1: Dot products

Let $u = (-3, 2)^T$ and $v = (4, 5)^T$. Compute $u \cdot v$ and decide whether the angle is acute, right, or obtuse.

Exercise 2: Three dimensions

Let $a = (1, -2, 4)^T$ and $b = (3, 0, -1)^T$. Find $a \cdot b$, $\|a\|$, $\|b\|$, and $\|a - b\|$.

Exercise 3: Unknown component

Find k so that $u = (2, k, 3)^T$ is orthogonal to $v = (-1, 4, 2)^T$.

Exercise 4: Angle between vectors

Find the exact value of $\cos \theta$ and then θ (to the nearest degree) for $u = (3, 1)^T$ and $v = (-2, 4)^T$.

Exercise 5: Projection

Find the projection of $u = (6, 1)^T$ onto $v = (2, -1)^T$. Verify that $u - \text{proj}_v(u)$ is orthogonal to v .

Exercise 6: Geometric reasoning

Suppose nonzero vectors p and q satisfy $p \cdot q = \|p\|\|q\|$. What can you conclude about their directions? Explain using the angle formula.

Part B · Cross products and geometry

Exercise 7: Basic cross product

For $a = (2, 1, -1)^T$ and $b = (0, 3, 2)^T$, compute $a \times b$ and $b \times a$. Verify the sign relation.

Exercise 8: Orthogonality check

Using your answer to Exercise 7, verify directly that $a \times b$ is orthogonal to both a and b .

Exercise 9: Parallelogram area

Find the area of the parallelogram with adjacent side vectors $u = (1, 2, 2)^T$ and $v = (3, -1, 0)^T$.

Exercise 10: Triangle area

The points are $P=(1,0,1)$, $Q=(3,1,0)$, and $R=(0,2,4)$. Find the area of triangle PQR.

Exercise 11: Parallel vectors

Find all real t for which $a = (1, t, 2)^T$ and $b = (3, 6, 6)^T$ have $a \times b = 0$. If no value exists, explain why.

Exercise 12: Challenge: volume

Let $a=(1,0,2)^T$, $b=(2,1,-1)^T$, $c=(0,3,1)^T$. Compute $a \cdot (b \times c)$. The absolute value is the volume of the parallelepiped. Interpret a zero value, if it occurs.

Part C · Matrices and basic operations

Exercise 13: Size, entries, rows, and columns

Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$. State its size. Find a_{12} and a_{23} . Write every row and every column as a vector, using column form for vectors.

Exercise 14: Matrix arithmetic

Let $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$. Compute $A+B$, $A-B$, $-2A$, and $3A-2B$.

Exercise 15: Compatibility

Let A be 4×3 , B be 3×2 , $x \in \mathbb{R}^2$, $y \in \mathbb{R}^3$, and $z \in \mathbb{R}^4$. Decide which of Ax , Ay , Az , Bx , By , and Bz are defined. State each output size.

Exercise 16: Diagonal action

Let $D = \text{diag}(-2, 3, 1)$ and $x = (4, -1, 5)^T$. Compute Dx . Describe in words what D does to each coordinate.

Exercise 17: Matrix-vector product

Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}$ and $x = (2, -1, 3)^T$. Compute Ax (a) as a linear combination of the columns and (b) using row dot products.

Exercise 18: Find the weights

Let $a_1 = (1, 2)^T$ and $a_2 = (3, -1)^T$. Find $x = (x_1, x_2)^T$ so that $x_1 a_1 + x_2 a_2 = (11, 1)^T$.

Exercise 19: Dependent columns

Let $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$. Describe all $x = (x_1, x_2)^T$ satisfying $Ax = 0$. Explain why the two columns are linearly dependent.

Exercise 20: Challenge: recover a transformation

A is a 3×2 matrix. You know $A(1, 0)^T = (2, -1, 4)^T$ and $A(0, 1)^T = (3, 2, 0)^T$. Write A and calculate $A(-2, 5)^T$ without first using row dot products.

Exercise 21: Challenge: a general identity

For any vectors u and v of the same dimension, expand $(u+v) \cdot (u-v)$. Use the result to prove that $u+v$ is orthogonal to $u-v$ exactly when $\|u\| = \|v\|$.

Self-check before Week 3

Without notes, make sure that you can answer each question.

1. What type of object does a dot product produce? What type does a cross product produce?
2. How can the sign of $u \cdot v$ classify an angle?
3. Why does a 2×3 matrix multiply a vector in $\mathbb{R}^3$, but not a vector in $\mathbb{R}^2$, on the right?
4. How do the entries of x become weights in Ax ?
5. Which Week 2 operations require equal dimensions, and which require vectors specifically in $\mathbb{R}^3$?

Reading: Strang, Introduction to Linear Algebra, 4th ed., Sections 1.2–1.3. Review Section 1.1 when linear combinations or independence are unclear.